Causal Analysis Report

BPI_Challenge_2013_closed_problems.xes.gz-im

Decision points:	16315
Choice sets:	10
Transitions fitted:	10
Significant (p<0.05):	1
Generated:	2026-06-22 07:27

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

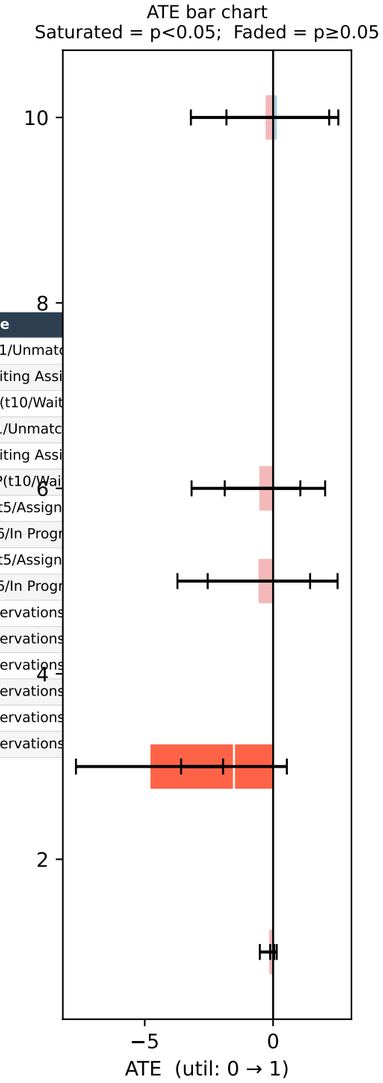
(no data)

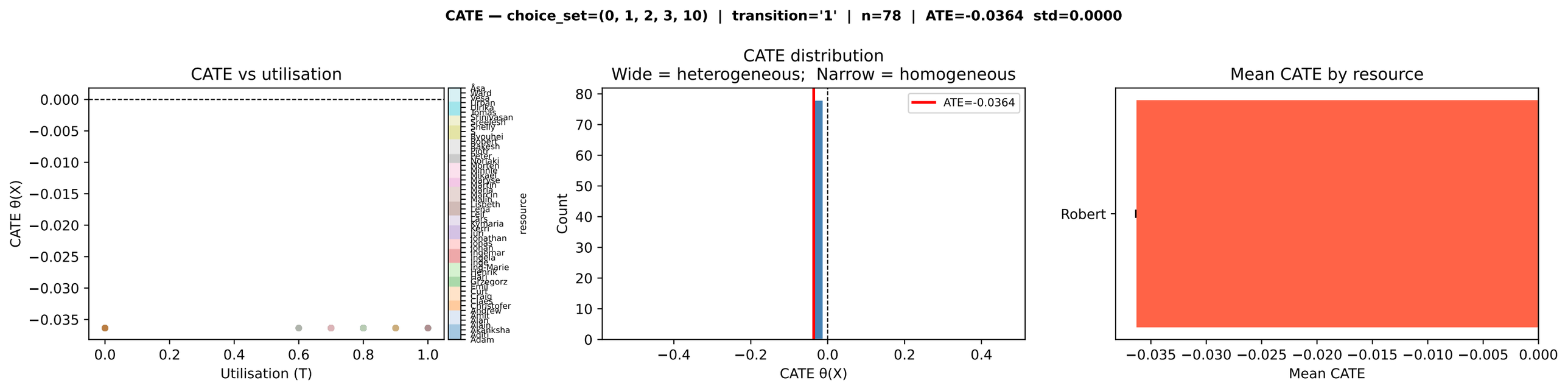
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

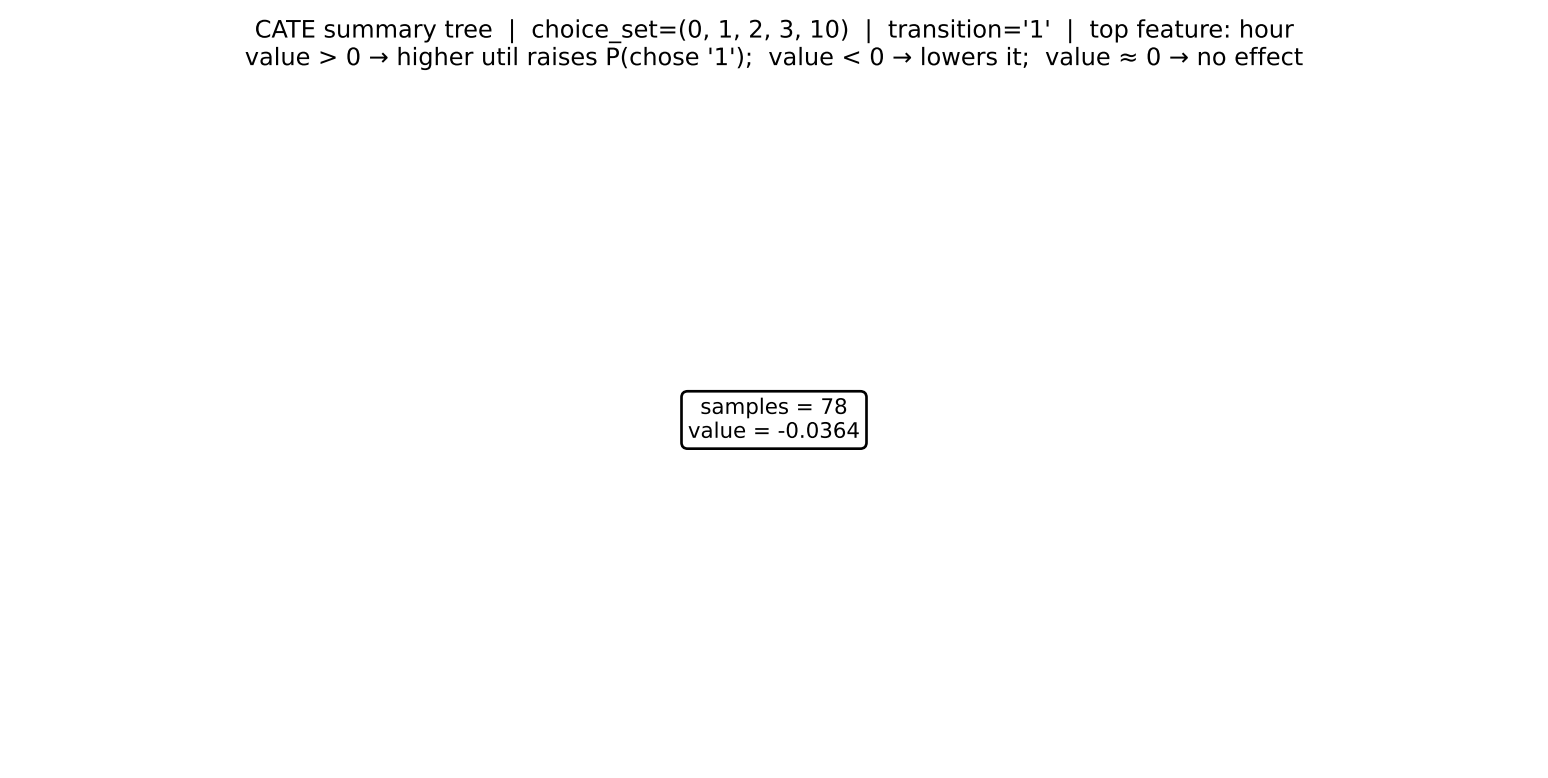
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	1	Unmatched_t1	Awaiting Assignment_t3, Wait_t10, t0_t0, t2_t2	-0.015608	0.110022	[-0.2313, 0.2000]	0.8872	False	θ=-0.0156: 1-unit ↑ util ↓ P(t1/Unmatched_t1) by 1.6%
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	3	Awaiting Assignment_t3	Unmatched_t1, Wait_t10, t0_t0, t2_t2	-1.739656	0.707644	[-3.1266, -0.3527]	0.014	True	θ=-1.7397: 1-unit ↑ util ↓ P(t3/Awaiting Assignment_t3) by 174.0%
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	10	Wait_t10	Awaiting Assignment_t3, Unmatched_t1, t0_t0, t2_t2	0.258668	0.532736	[-0.7855, 1.3028]	0.6273	False	θ=+0.2587: 1-unit ↑ util ↑ P(t10/Wait_t10) by 25.9%
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	1	Unmatched_t1	Awaiting Assignment_t3, Wait_t10	-0.147871	0.059778	[-0.2650, -0.0307]	0.0134	True	θ=-0.1479: 1-unit ↑ util ↓ P(t1/Unmatched_t1) by 14.8%
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	3	Awaiting Assignment_t3	Unmatched_t1, Wait_t10	-0.203195	0.25626	[-0.7055, 0.2991]	0.4278	False	θ=-0.2032: 1-unit ↑ util ↓ P(t3/Awaiting Assignment_t3) by 20.3%
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	10	Wait_t10	Awaiting Assignment_t3, Unmatched_t1	0.358463	0.228651	[-0.0897, 0.8066]	0.1169	False	θ=+0.3585: 1-unit ↑ util ↑ P(t10/Wait_t10) by 35.8%
(4, 5, 6)	('τ_t4', 'Assigned_t5', 'In Progress_t6')	809	5, 6	5	Assigned_t5	In Progress_t6	-0.530731	0.323274	[-1.1643, 0.1029]	0.1006	False	θ=-0.5307: 1-unit ↑ util ↓ P(t5/Assigned_t5) by 53.1%
(4, 5, 6)	('τ_t4', 'Assigned_t5', 'In Progress_t6')	809	5, 6	6	In Progress_t6	Assigned_t5	-0.254106	0.261441	[-0.7665, 0.2583]	0.3311	False	θ=-0.2541: 1-unit ↑ util ↓ P(t6/In Progress_t6) by 25.4%
(5, 6)	('Assigned_t5', 'In Progress_t6')	2538	5, 6	5	Assigned_t5	In Progress_t6	-0.559749	0.280985	[-1.1105, -0.0090]	0.0464	True	θ=-0.5597: 1-unit ↑ util ↓ P(t5/Assigned_t5) by 56.0%
(5, 6)	('Assigned_t5', 'In Progress_t6')	2538	5, 6	6	In Progress_t6	Assigned_t5	-0.733121	0.230804	[-1.1855, -0.2807]	0.0015	True	θ=-0.7331: 1-unit ↑ util ↓ P(t6/In Progress_t6) by 73.3%
(6, 12)	('In Progress_t6', 'Closed_t12')	333	6	—	—	—	nan	nan	—	nan	—	too few observations
(7, 8)	('τ_t7', 'Cancelled_t8')	3	8	—	—	—	nan	nan	—	nan	—	too few observations
(7, 8, 12)	('τ_t7', 'Cancelled_t8', 'Closed_t12')	164	12	—	—	—	nan	nan	—	nan	—	too few observations
(9, 11, 12)	('τ_t9', 'τ_t11', 'Closed_t12')	1102	12	—	—	—	nan	nan	—	nan	—	too few observations
(11, 12)	('τ_t11', 'Closed_t12')	298	12	—	—	—	nan	nan	—	nan	—	too few observations
(13, 14)		0		—	—	—	nan	nan	—	nan	—	too few observations

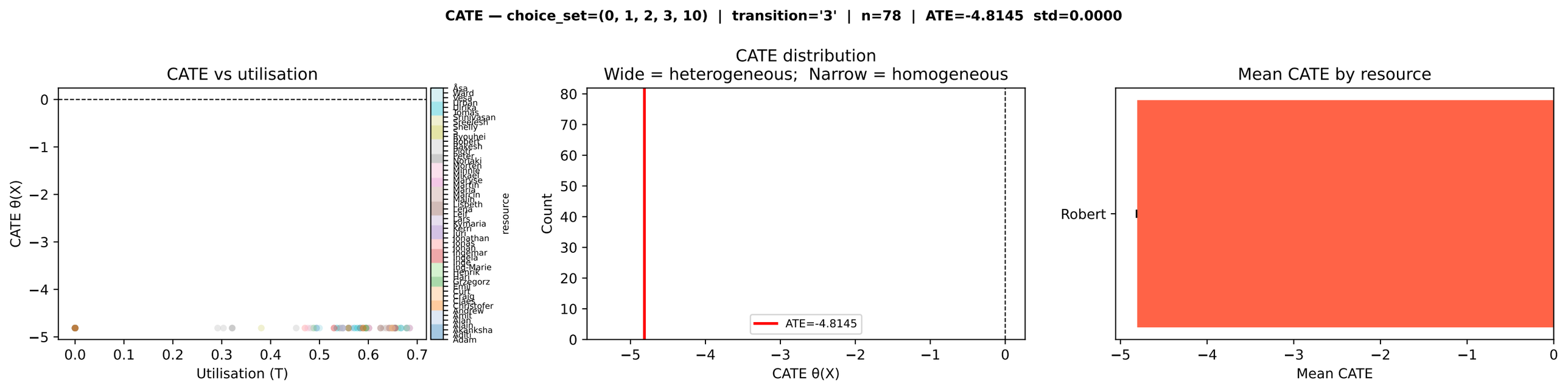
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

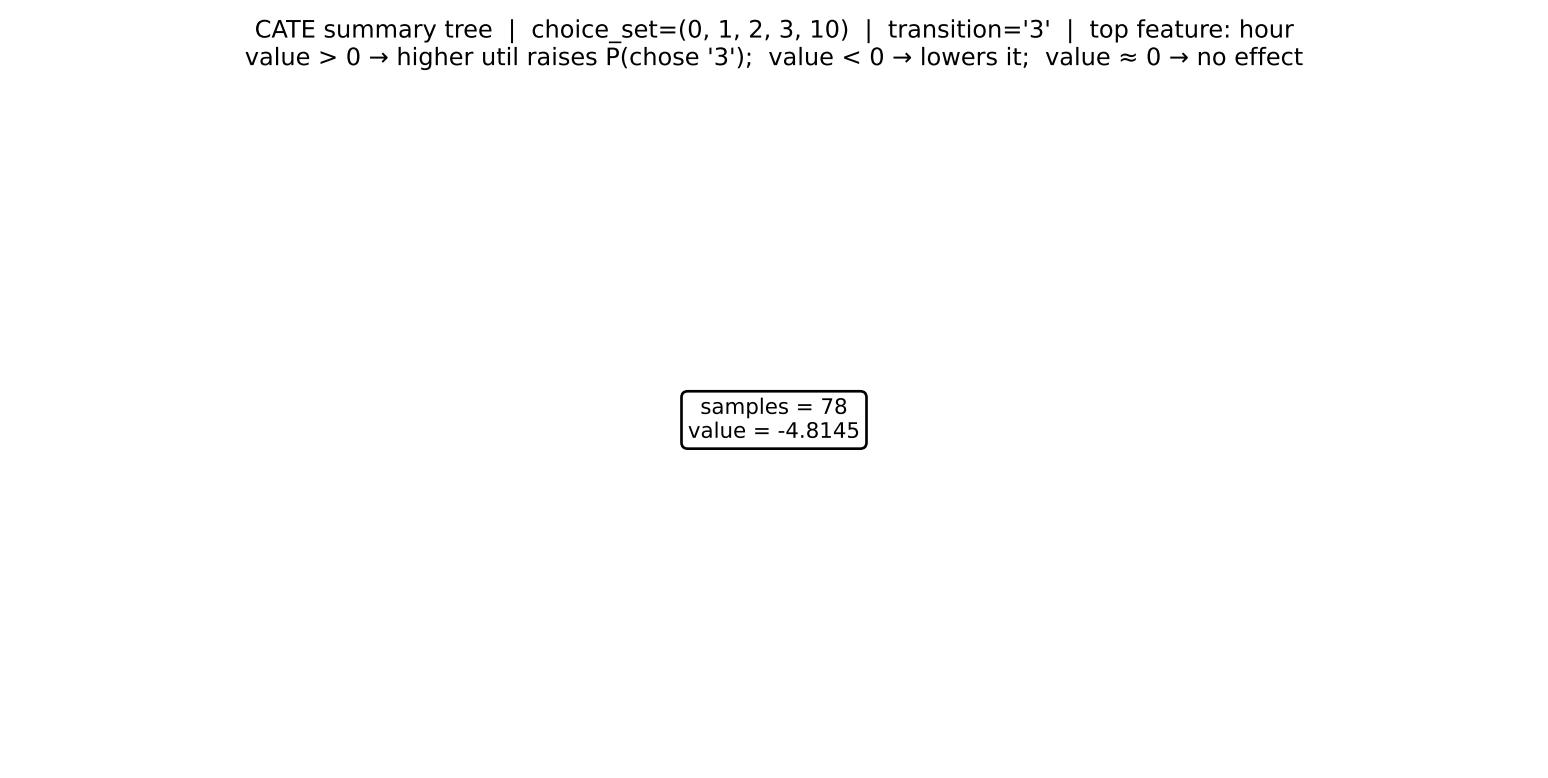
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	1	Unmatched_t1	Awaiting Assignment_t3, Wait_t10	-0.036365	[-0.1151, 0.0423]	0.3651	False	ATE=-0.0364: 0→1 util ↓ P(t1/Unmatc
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	3	Awaiting Assignment_t3	Unmatched_t1, Wait_t10	-4.814491	[-7.6796, -1.9494]	0.001	True	ATE=-4.8145: 0→1 util ↓ P(t3/Awaiting Assi
(0, 1, 2, 3, 10)	('τ_t0', 'Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	78	1, 3, 10	10	Wait_t10	Awaiting Assignment_t3, Unmatched_t1	-0.330833	[-3.2018, 2.5402]	0.8213	False	ATE=-0.3308: 0→1 util ↓ P(t10/Wait
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	1	Unmatched_t1	Awaiting Assignment_t3, Wait_t10	-0.187656	[-0.5167, 0.1414]	0.2636	False	ATE=-0.1877: 0→1 util ↓ P(t1/Unmatc
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	3	Awaiting Assignment_t3	Unmatched_t1, Wait_t10	-1.524041	[-3.5841, 0.5360]	0.1471	False	ATE=-1.5240: 0→1 util ↓ P(t3/Awaiting Assi
(1, 2, 3, 10)	('Unmatched_t1', 'τ_t2', 'Awaiting Assignment_t3', 'Wait_t10')	1268	1, 3, 10	10	Wait_t10	Awaiting Assignment_t3, Unmatched_t1	0.183434	[-1.8174, 2.1843]	0.8574	False	ATE=+0.1834: 0→1 util ↑ P(t10/Wai
(4, 5, 6)	('τ_t4', 'Assigned_t5', 'In Progress_t6')	809	5, 6	5	Assigned_t5	In Progress_t6	-0.553631	[-2.5488, 1.4415]	0.5865	False	ATE=-0.5536: 0→1 util ↓ P(t5/Assign
(4, 5, 6)	('τ_t4', 'Assigned_t5', 'In Progress_t6')	809	5, 6	6	In Progress_t6	Assigned_t5	-0.411546	[-1.8840, 1.0609]	0.5838	False	ATE=-0.4115: 0→1 util ↓ P(t6/In Progr
(5, 6)	('Assigned_t5', 'In Progress_t6')	2538	5, 6	5	Assigned_t5	In Progress_t6	-0.607617	[-3.7264, 2.5112]	0.7026	False	ATE=-0.6076: 0→1 util ↓ P(t5/Assign
(5, 6)	('Assigned_t5', 'In Progress_t6')	2538	5, 6	6	In Progress_t6	Assigned_t5	-0.574091	[-3.1755, 2.0273]	0.6654	False	ATE=-0.5741: 0→1 util ↓ P(t6/In Progr
(6, 12)	('In Progress_t6', 'Closed_t12')	333	6	—	—	—	nan	—	nan	—	too few observations
(7, 8)	('τ_t7', 'Cancelled_t8')	3	8	—	—	—	nan	—	nan	—	too few observations
(7, 8, 12)	('τ_t7', 'Cancelled_t8', 'Closed_t12')	164	12	—	—	—	nan	—	nan	—	too few observations
(9, 11, 12)	('τ_t9', 'τ_t11', 'Closed_t12')	1102	12	—	—	—	nan	—	nan	—	too few observations
(11, 12)	('τ_t11', 'Closed_t12')	298	12	—	—	—	nan	—	nan	—	too few observations
(13, 14)		0		—	—	—	nan	—	nan	—	too few observations

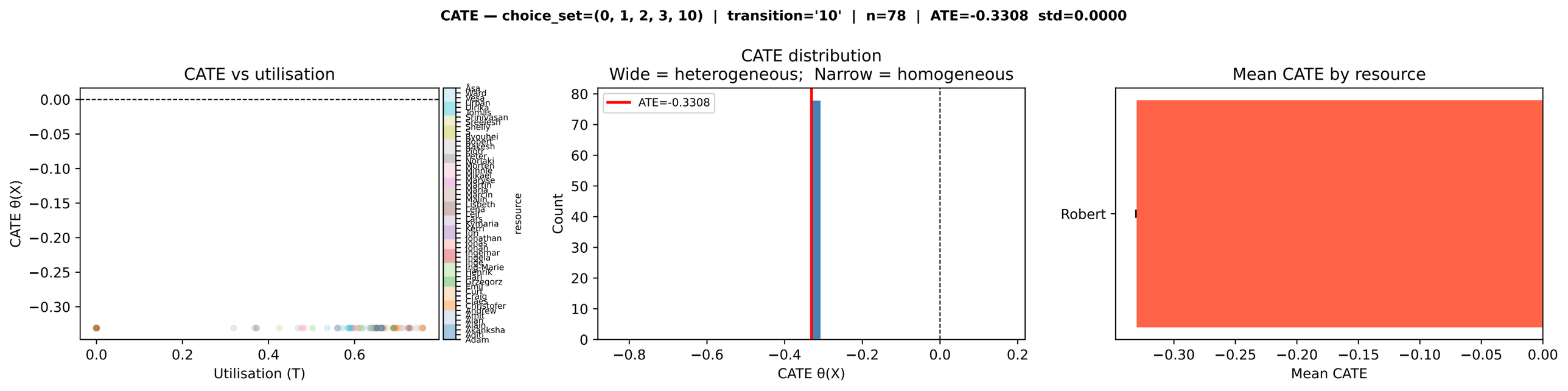


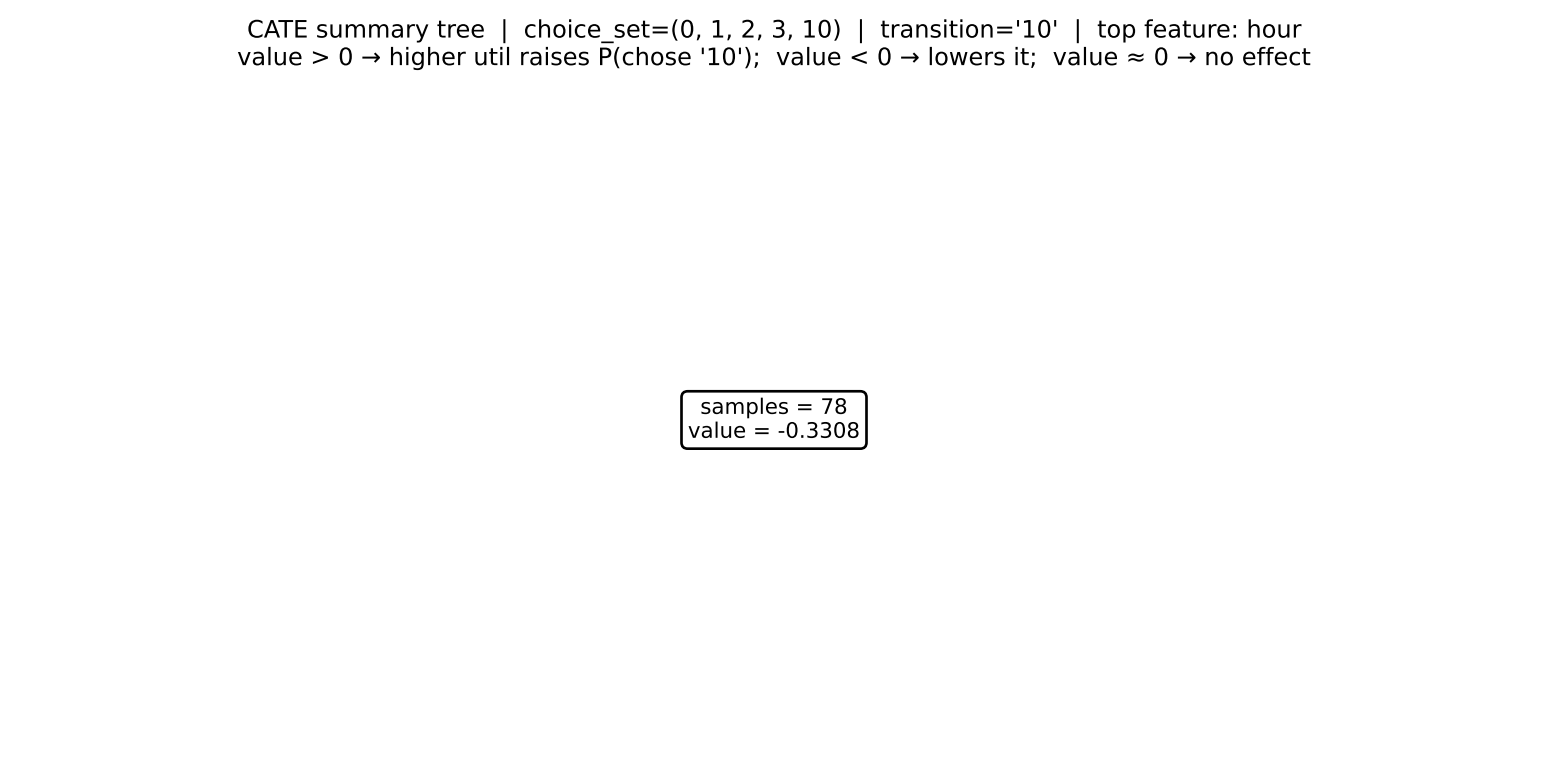

CATE — choice_set=(0, 1, 2, 3, 10) | transition='1' | n=78 | ATE=-0.0364 std=0.0000
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.0364
Mean CATE by resource
Robert
Mean CATE

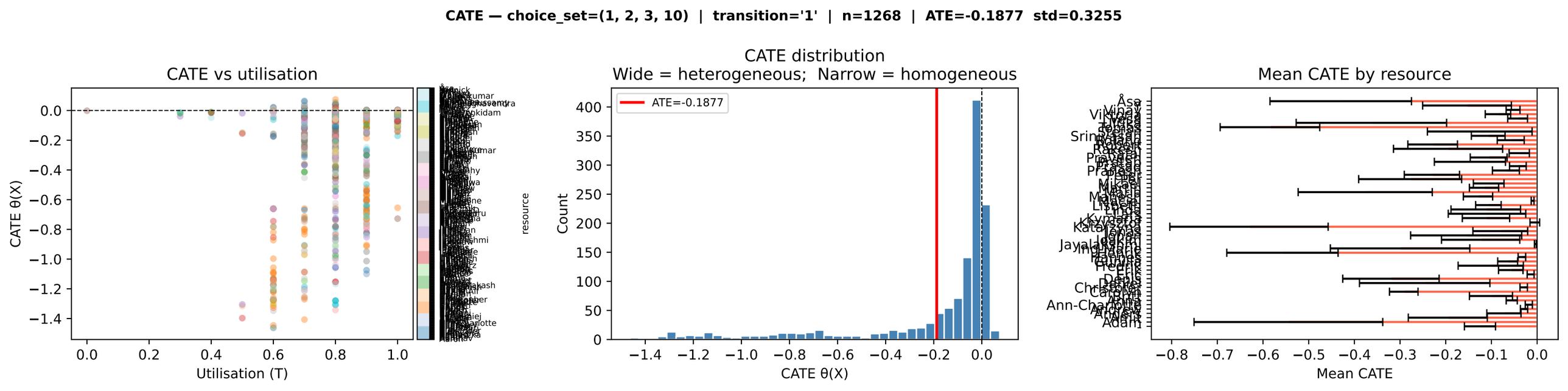

CATE summary tree | choice_set=(0, 1, 2, 3, 10) | transition='1' | top feature: hour
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
samples = 78
value = -0.0364

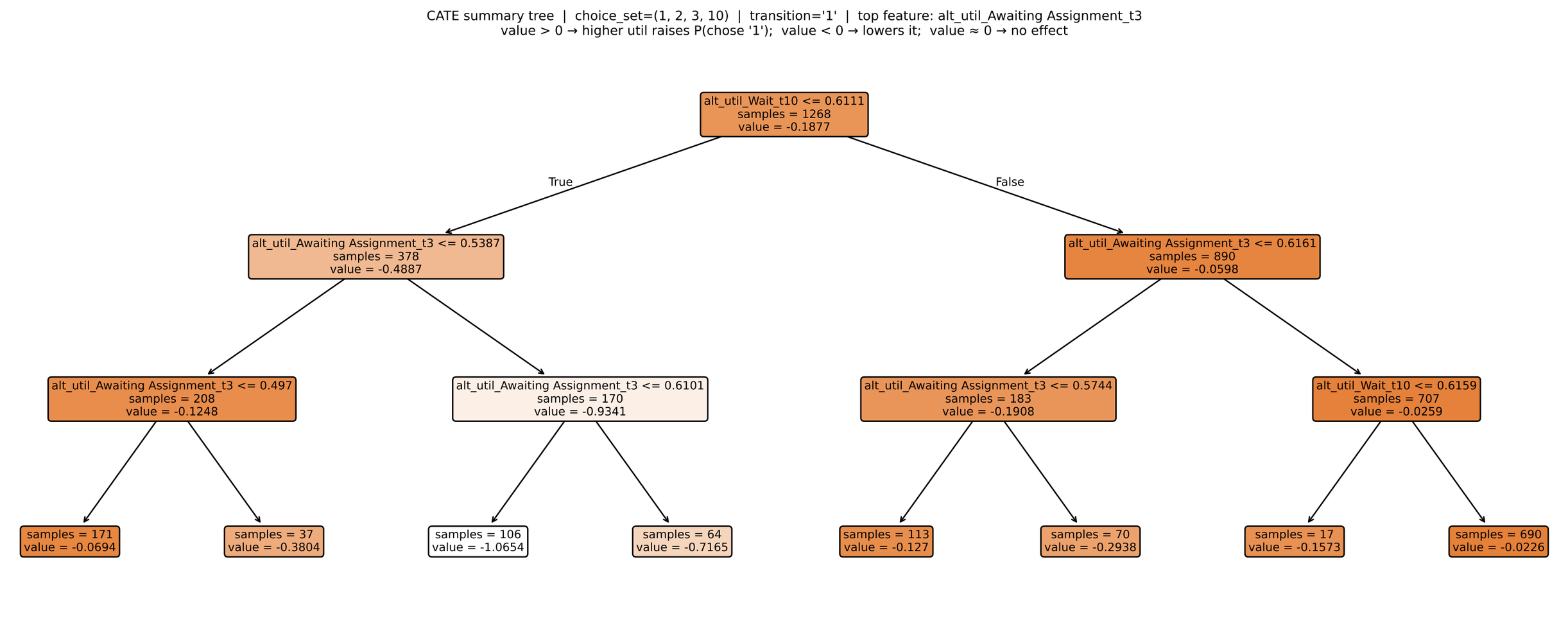

CATE — choice_set=(0, 1, 2, 3, 10) | transition='3' | n=78 | ATE=-4.8145 std=0.0000
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
ATE=-4.8145
CATE θ(X)
Mean CATE by resource
Robert
Mean CATE

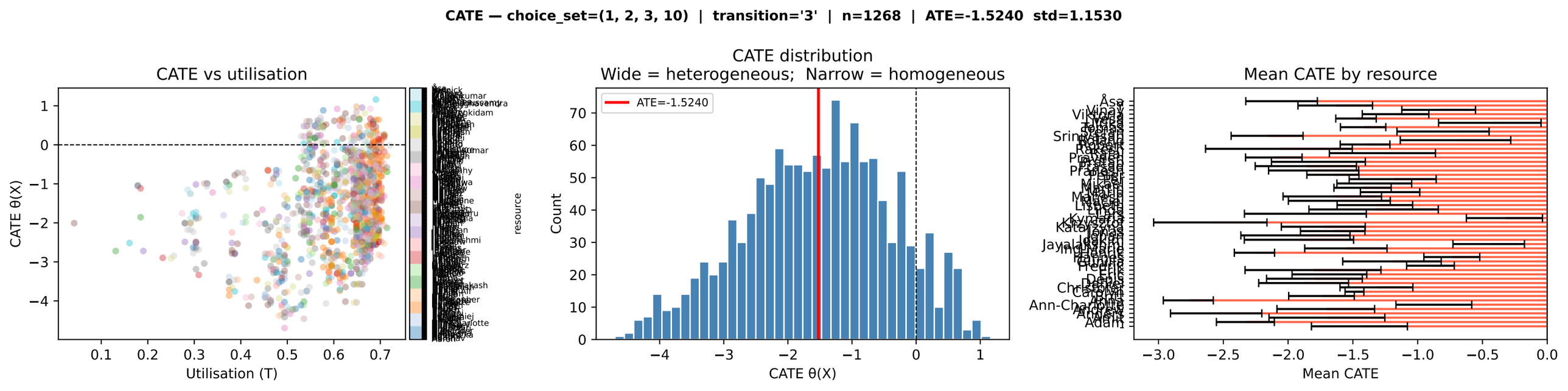

CATE summary tree | choice_set=(0, 1, 2, 3, 10) | transition='3' | top feature: hour
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
samples = 78
value = -4.8145

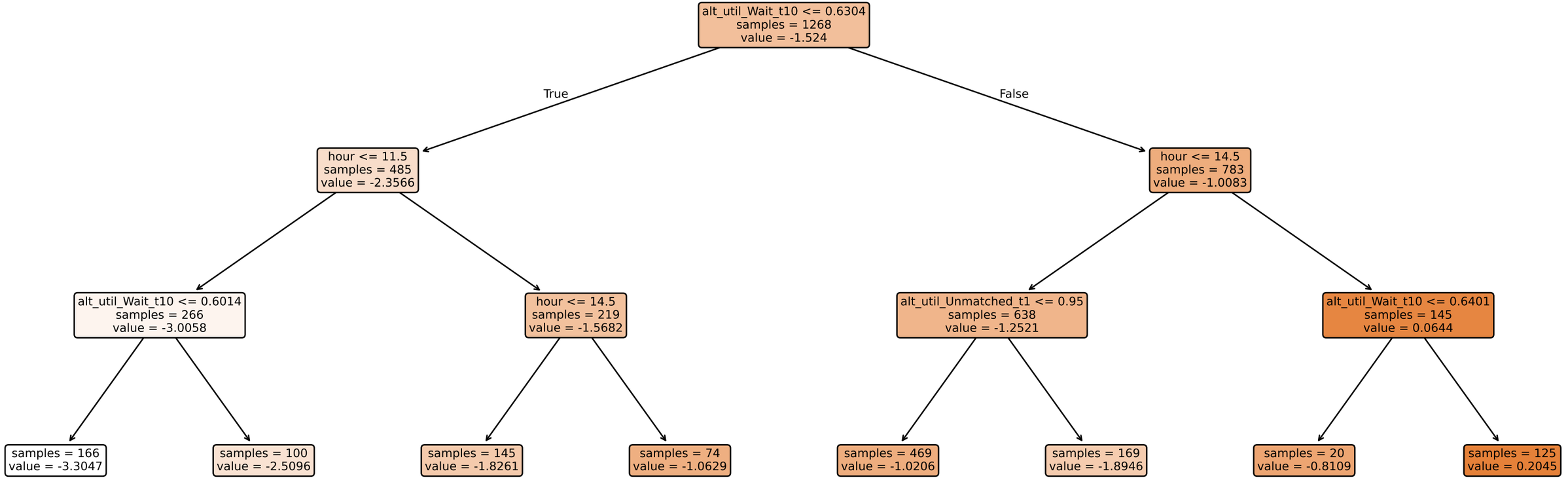

CATE — choice_set=(0, 1, 2, 3, 10) | transition='10' | n=78 | ATE=-0.3308 std=0.0000
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3308
Count
CATE θ(X)
Mean CATE by resource
Robert
Mean CATE

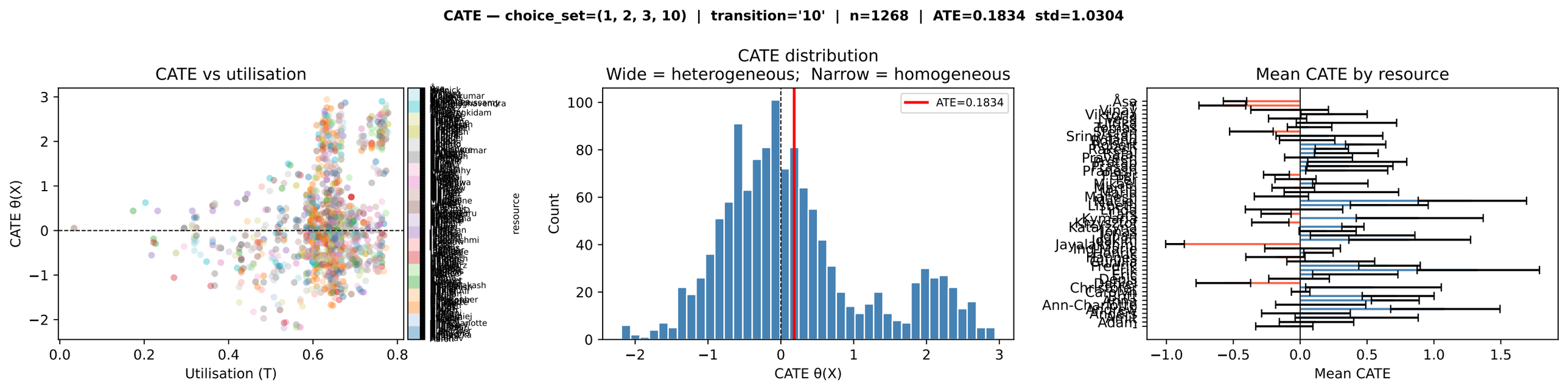

CATE summary tree | choice_set=(0, 1, 2, 3, 10) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
samples = 78
value = -0.3308

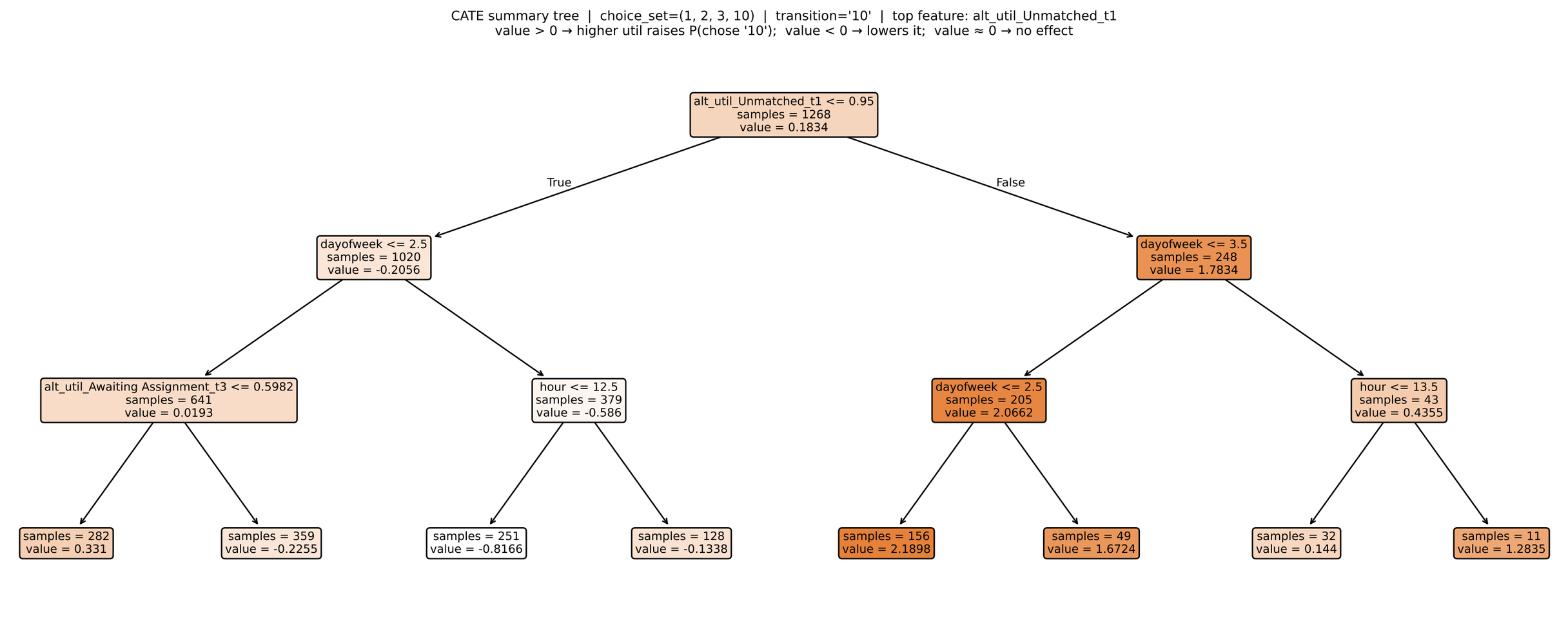

CATE — choice_set=(1, 2, 3, 10) | transition='1' | n=1268 | ATE=-0.1877 std=0.3255
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1877
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

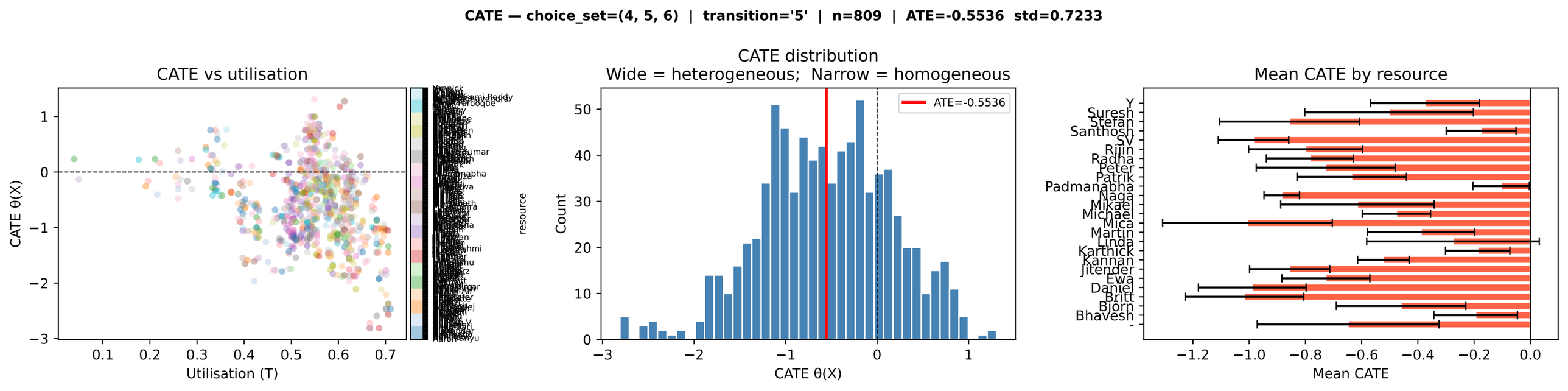

CATE summary tree | choice_set=(1, 2, 3, 10) | transition='1' | top feature: alt_util_Awaiting Assignment_t3
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Wait_t10 <= 0.6111
samples = 1268
value = -0.1877
True
False
alt_util_Awaiting Assignment_t3 <= 0.5387
samples = 378
value = -0.4887
alt_util_Awaiting Assignment_t3 <= 0.6161
samples = 890
value = -0.0598
alt_util_Awaiting Assignment_t3 <= 0.497
samples = 208
value = -0.1248
alt_util_Awaiting Assignment_t3 <= 0.6101
samples = 170
value = -0.9341
alt_util_Awaiting Assignment_t3 <= 0.5744
samples = 183
value = -0.1908
alt_util_Wait_t10 <= 0.6159
samples = 707
value = -0.0259
samples = 171
value = -0.0694
samples = 37
value = -0.3804
samples = 106
value = -1.0654
samples = 64
value = -0.7165
samples = 113
value = -0.127
samples = 70
value = -0.2938
samples = 17
value = -0.1573
samples = 690
value = -0.0226

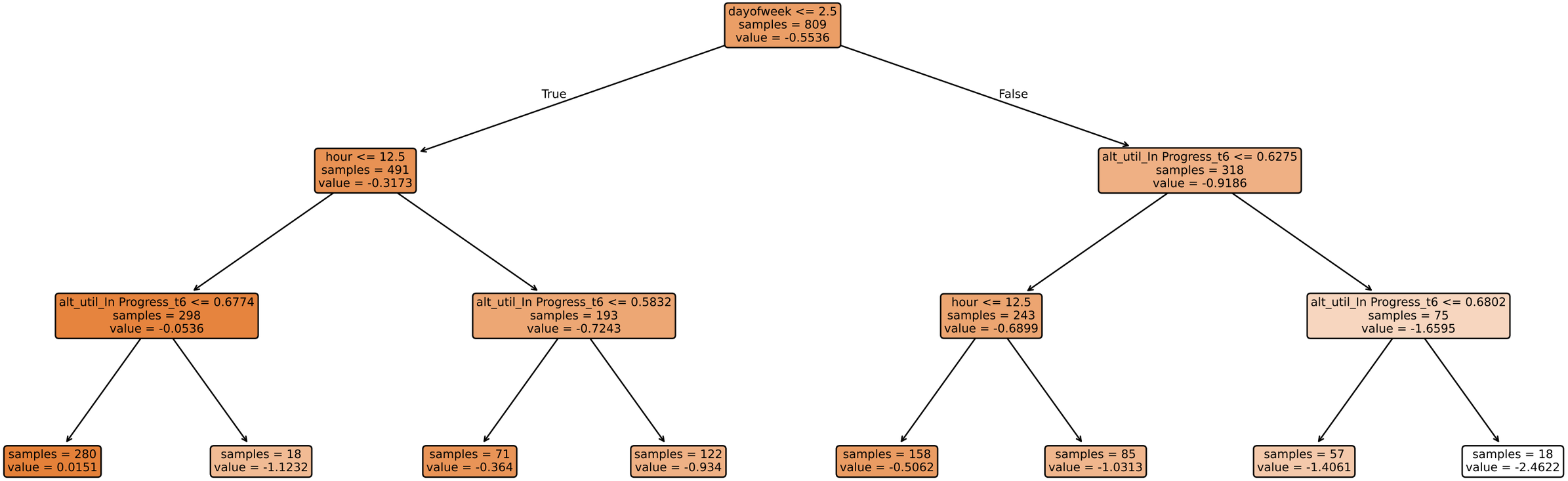

CATE — choice_set=(1, 2, 3, 10) | transition='3' | n=1268 | ATE=-1.5240 std=1.1530
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.5240
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 10) | transition='3' | top feature: alt_util_Wait_t10
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Wait_t10 <= 0.6304
samples = 1268
value = -1.524
True
False
hour <= 11.5
samples = 485
value = -2.3566
hour <= 14.5
samples = 783
value = -1.0083
alt_util_Wait_t10 <= 0.6014
samples = 266
value = -3.0058
hour <= 14.5
samples = 219
value = -1.5682
alt_util_Unmatched_t1 <= 0.95
samples = 638
value = -1.2521
alt_util_Wait_t10 <= 0.6401
samples = 145
value = 0.0644
samples = 166
value = -3.3047
samples = 100
value = -2.5096
samples = 145
value = -1.8261
samples = 74
value = -1.0629
samples = 469
value = -1.0206
samples = 169
value = -1.8946
samples = 20
value = -0.8109
samples = 125
value = 0.2045

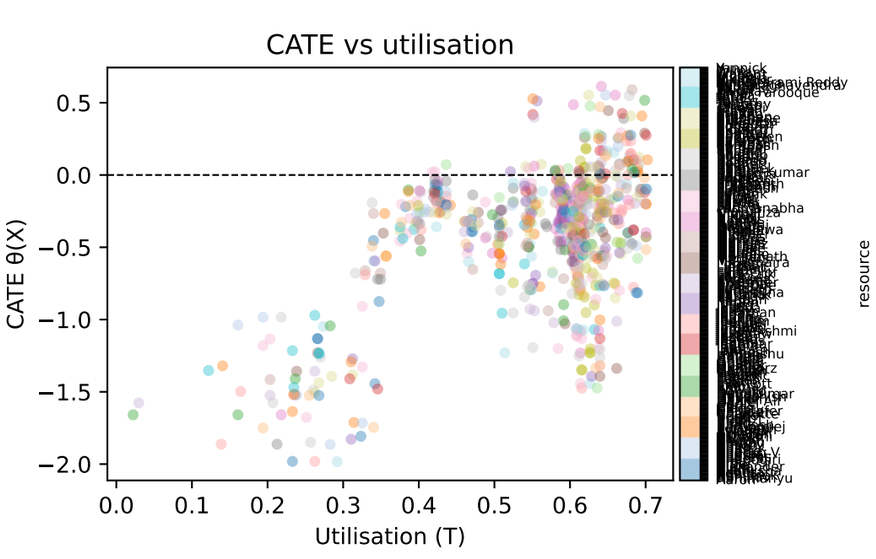

CATE — choice_set=(1, 2, 3, 10) | transition='10' | n=1268 | ATE=0.1834 std=1.0304
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=0.1834
Mean CATE by resource
Mean CATE

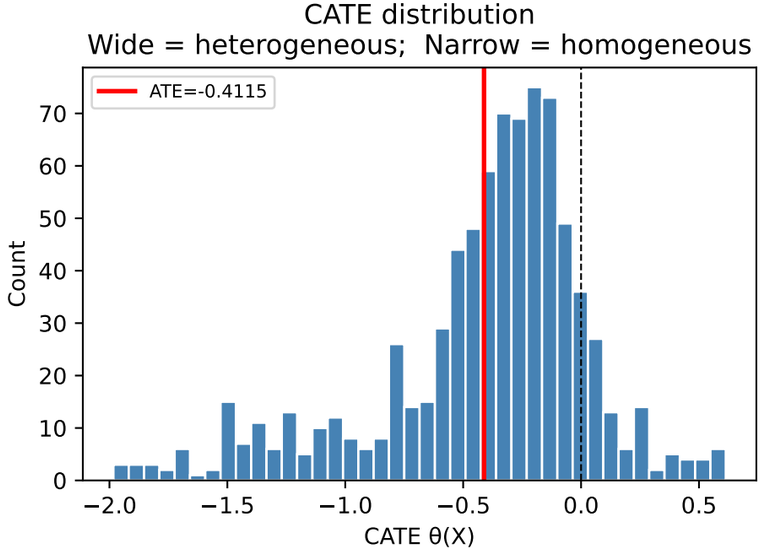

CATE summary tree | choice_set=(1, 2, 3, 10) | transition='10' | top feature: alt_util_Unmatched_t1
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Unmatched_t1 <= 0.95
samples = 1268
value = 0.1834
True
False
dayofweek <= 2.5
samples = 1020
value = -0.2056
dayofweek <= 3.5
samples = 248
value = 1.7834
alt_util_Awaiting Assignment_t3 <= 0.5982
samples = 641
value = 0.0193
hour <= 12.5
samples = 379
value = -0.586
dayofweek <= 2.5
samples = 205
value = 2.0662
hour <= 13.5
samples = 43
value = 0.4355
samples = 282
value = 0.331
samples = 359
value = -0.2255
samples = 251
value = -0.8166
samples = 128
value = -0.1338
samples = 156
value = 2.1898
samples = 49
value = 1.6724
samples = 32
value = 0.144
samples = 11
value = 1.2835

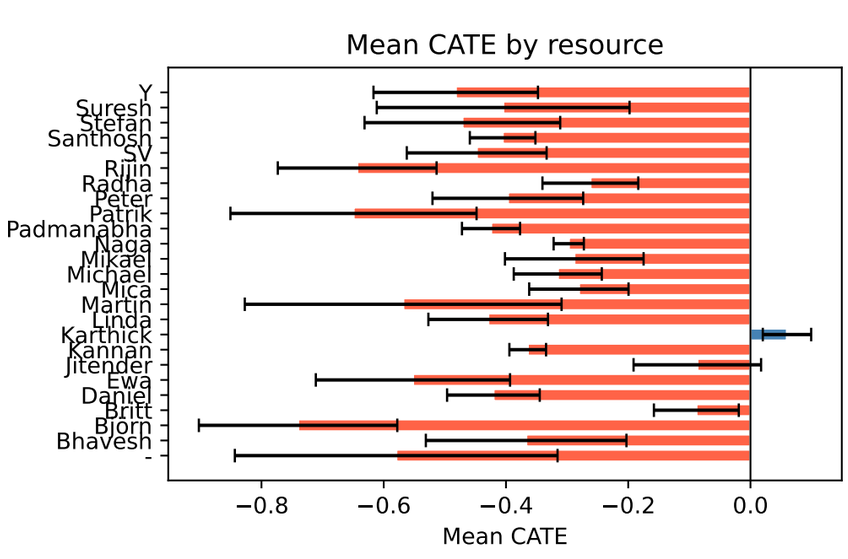

CATE — choice_set=(4, 5, 6) | transition='5' | n=809 | ATE=-0.5536 std=0.7233
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5536
Count
CATE θ(X)
Mean CATE by resource
Y
Suresh
Stefan
Santhosh
SV
Rijin
Radha
Peter
Patrik
Padmanabha
Naga
Mikael
Michael
Mica
Martin
Linda
Karthick
Kannan
Jitender
Ewa
Daniel
Britt
Bjorn
Bhavesh
-
Mean CATE

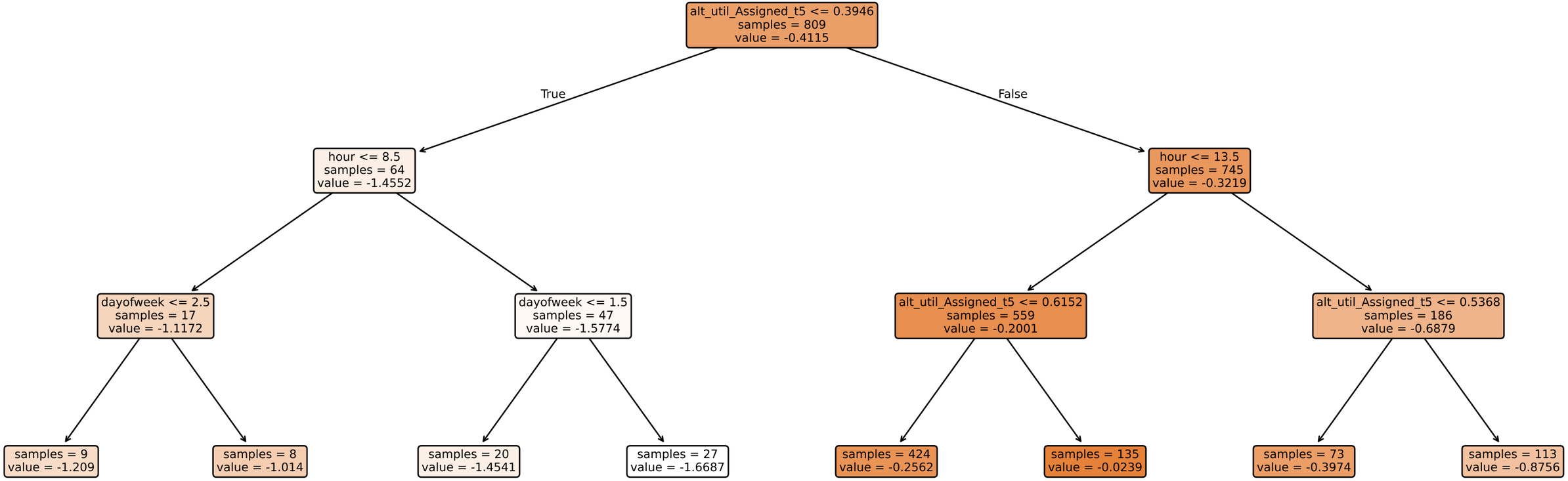

CATE summary tree | choice_set=(4, 5, 6) | transition='5' | top feature: alt_util_ln Progress_t6
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 809
value = -0.5536
True
False
hour <= 12.5
samples = 491
value = -0.3173
alt_util_ln Progress_t6 <= 0.6275
samples = 318
value = -0.9186
alt_util_ln Progress_t6 <= 0.6774
samples = 298
value = -0.0536
alt_util_ln Progress_t6 <= 0.5832
samples = 193
value = -0.7243
hour <= 12.5
samples = 243
value = -0.6899
alt_util_ln Progress_t6 <= 0.6802
samples = 75
value = -1.6595
samples = 280
value = 0.0151
samples = 18
value = -1.1232
samples = 71
value = -0.364
samples = 122
value = -0.934
samples = 158
value = -0.5062
samples = 85
value = -1.0313
samples = 57
value = -1.4061
samples = 18
value = -2.4622

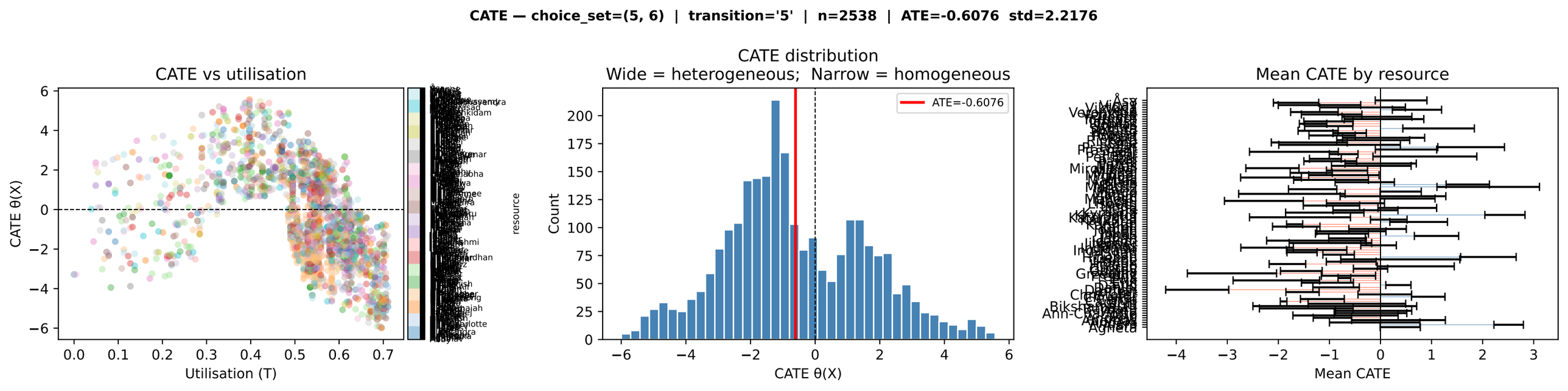
CATE — choice_set=(4, 5, 6) | transition='6' | n=809 | ATE=-0.4115 std=0.4574

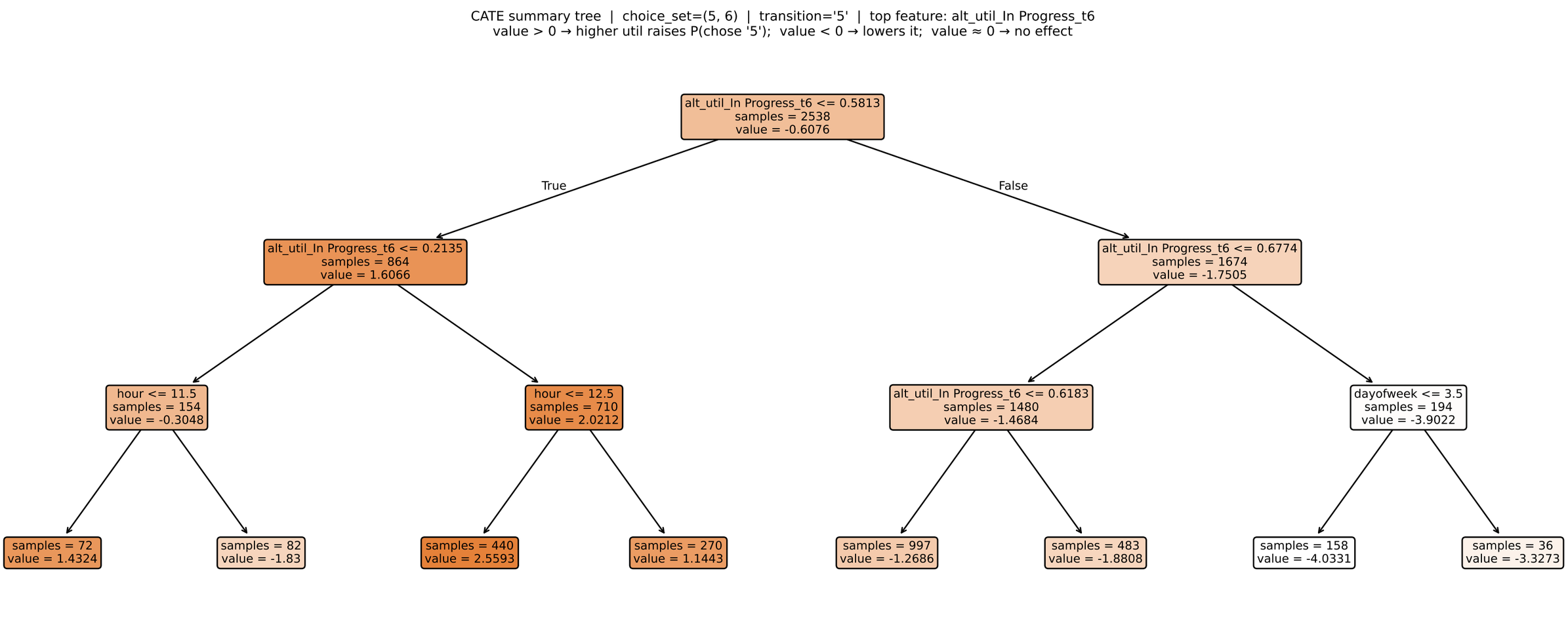


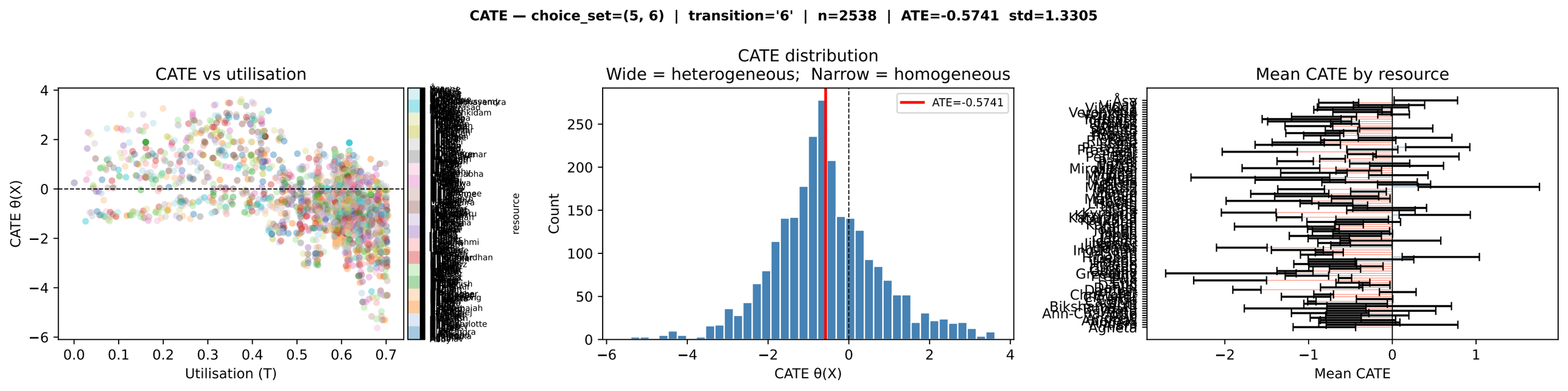


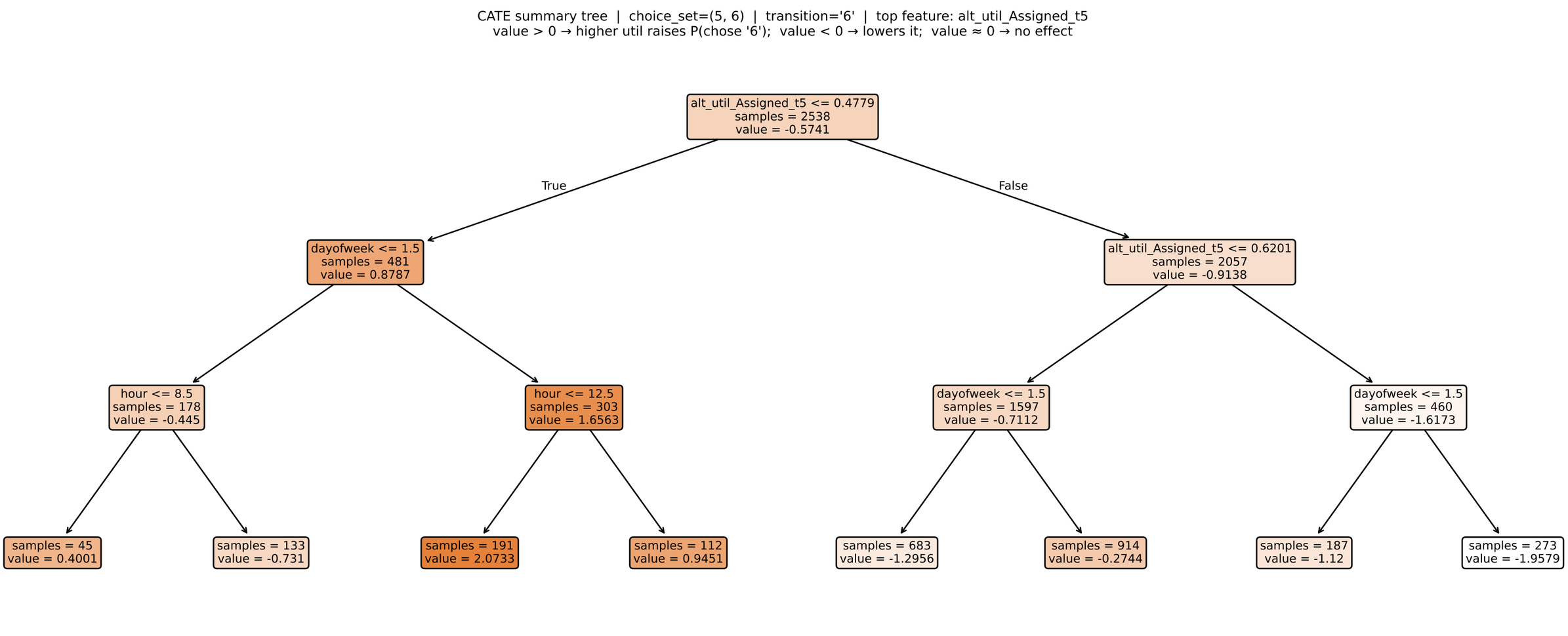



CATE summary tree | choice_set=(4, 5, 6) | transition='6' | top feature: alt_util_Assigned_t5
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t5 <= 0.3946
samples = 809
value = -0.4115
True
False
hour <= 8.5
samples = 64
value = -1.4552
hour <= 13.5
samples = 745
value = -0.3219
dayofweek <= 2.5
samples = 17
value = -1.1172
dayofweek <= 1.5
samples = 47
value = -1.5774
alt_util_Assigned_t5 <= 0.6152
samples = 559
value = -0.2001
alt_util_Assigned_t5 <= 0.5368
samples = 186
value = -0.6879
samples = 9
value = -1.209
samples = 8
value = -1.014
samples = 20
value = -1.4541
samples = 27
value = -1.6687
samples = 424
value = -0.2562
samples = 135
value = -0.0239
samples = 73
value = -0.3974
samples = 113
value = -0.8756


CATE — choice_set=(5, 6) | transition='5' | n=2538 | ATE=-0.6076 std=2.2176
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.6076
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(5, 6) | transition='5' | top feature: alt_util_ln Progress_t6
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t6 <= 0.5813
samples = 2538
value = -0.6076
True
False
alt_util_ln Progress_t6 <= 0.2135
samples = 864
value = 1.6066
alt_util_ln Progress_t6 <= 0.6774
samples = 1674
value = -1.7505
hour <= 11.5
samples = 154
value = -0.3048
hour <= 12.5
samples = 710
value = 2.0212
alt_util_ln Progress_t6 <= 0.6183
samples = 1480
value = -1.4684
dayofweek <= 3.5
samples = 194
value = -3.9022
samples = 72
value = 1.4324
samples = 82
value = -1.83
samples = 440
value = 2.5593
samples = 270
value = 1.1443
samples = 997
value = -1.2686
samples = 483
value = -1.8808
samples = 158
value = -4.0331
samples = 36
value = -3.3273


CATE — choice_set=(5, 6) | transition='6' | n=2538 | ATE=-0.5741 std=1.3305
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5741
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(5, 6) | transition='6' | top feature: alt_util_Assigned_t5
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t5 <= 0.4779
samples = 2538
value = -0.5741
True
False
dayofweek <= 1.5
samples = 481
value = 0.8787
alt_util_Assigned_t5 <= 0.6201
samples = 2057
value = -0.9138
hour <= 8.5
samples = 178
value = -0.445
hour <= 12.5
samples = 303
value = 1.6563
dayofweek <= 1.5
samples = 1597
value = -0.7112
dayofweek <= 1.5
samples = 460
value = -1.6173
samples = 45
value = 0.4001
samples = 133
value = -0.731
samples = 191
value = 2.0733
samples = 112
value = 0.9451
samples = 683
value = -1.2956
samples = 914
value = -0.2744
samples = 187
value = -1.12
samples = 273
value = -1.9579

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(0, 1, 2, 3, 10)	1	78	-0.036365	0.0	-0.036365	-0.036365	-0.036365	-0.036365	-0.036365	0.0	100.0	—
(0, 1, 2, 3, 10)	3	78	-4.814491	0.0	-4.814491	-4.814491	-4.814491	-4.814491	-4.814491	0.0	100.0	—
(0, 1, 2, 3, 10)	10	78	-0.330833	0.0	-0.330833	-0.330833	-0.330833	-0.330833	-0.330833	0.0	100.0	—
(1, 2, 3, 10)	1	1268	-0.187656	0.325467	-1.463952	-0.170594	-0.037476	-0.00648	0.074299	12.0	88.0	alt_util_Awaiting Assignment_t3
(1, 2, 3, 10)	3	1268	-1.524041	1.153029	-4.696456	-2.339984	-1.47009	-0.691247	1.16159	9.7	90.3	alt_util_Wait_t10
(1, 2, 3, 10)	10	1268	0.183434	1.030387	-2.18938	-0.54385	-0.028639	0.601253	2.942138	48.5	51.5	alt_util_Unmatched_t1
(4, 5, 6)	5	809	-0.553631	0.7233	-2.80991	-1.064539	-0.554923	-0.047678	1.305418	23.4	76.6	alt_util_In Progress_t6
(4, 5, 6)	6	809	-0.411546	0.457359	-1.983136	-0.558693	-0.320976	-0.126553	0.613506	12.7	87.3	alt_util_Assigned_t5
(5, 6)	5	2538	-0.607617	2.217558	-5.99637	-2.095278	-0.978971	1.113321	5.568906	35.7	64.3	alt_util_In Progress_t6
(5, 6)	6	2538	-0.574091	1.330548	-5.632664	-1.368591	-0.683952	0.156064	3.623148	28.5	71.5	alt_util_Assigned_t5

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md